Hospitalization and In-Hospital Mortality Trends for Primary Malignant Brain Tumors in Ecuador: A 10-Year Population Study

Mónica Daniela Barreiro-Linzan[1,4*], Karime Montes-Escobar[2,3], Purificación Vicente Galindo[4], Aline Siteneski[5,6]

[1]Departamento de Matemáticas y Estadística, Facultad de Ciencias Básicas, Universidad Técnica de Manabí. Portoviejo. Ecuador

[2]Laboratorio de Funcionamiento de Agroecosistemas y Cambio Climático – FAGROCLIM, Departamento de Ciencias Agrícolas, Facultad de Ingeniería Agrícola. Universidad Técnica de Manabí. Portoviejo. Ecuador.

[3]Departamento de Formación y Desarrollo Científico en Ingeniería, Facultad de Ingeniería, Ciencia y Tecnología, Universidad Bernardo O'Higgins, Chile

[4]Department of Statistics, University of Salamanca, 37008 Salamanca, Spain

[5]Carrera de Medicina, Facultad de Ciencias de la Salud, Universidad Técnica de Manabí, Portoviejo, Manabí, Ecuador.

*** Corresponding author**

Mónica Daniela Barreiro Linzan

Departamento de Matemáticas y Estadística, Facultad de Ciencias Básicas, Universidad Técnica de Manabí. Portoviejo. Ecuador

Phone: 00 (593) 994 607 405

E-mail address: idu075591@usal.es, shachinaa03@hotmail.com

Abstract

Primary malignant brain tumors pose a major public health challenge in low- and middle-income countries, yet long-term population-based data from the Andean region remain scarce. This study evaluates 10-year nationwide trends, anatomical distributions, and determinants of hospitalization and in-hospital mortality in Ecuador (2015–2024). Nationwide retrospective analysis of hospital discharge records registries standardized to the 2022 Ecuadorian Census. Age-adjusted hospitalization (AAHR) and in-hospital mortality rates (AAIMR) per 100,000 population were calculated. Adjusted odds ratios (aOR) and relative risk ratios ($aRRR$) were estimated using multivariate logistic models. The overall 10-year mean AAHR was 4.859 per 100,000, while the AAIMR stood at 0.461 per 100,000. The overall in-hospital case-fatality proportion was 9.61%, while the 10-year mean AAIMR was 0.461 per 100,000 population.

Pediatric patients ($\leq$17 years) concentrated 43.2% of infratentorial/midline lesions ($aRRR = 2.424, p = 0.001$), whereas supratentorial tumors predominated in adults. A drop in total AAHR occurred during the COVID-19 pandemic in 2020 (5.322 per 100,000). Coding precision improved significantly by 2024, as unspecified tumor AAHR dropped to 0.000 per 100,000. Multivariate regression revealed that older age ($aOR = 0.457$ for $\geq$80 years, $p = 0.001$) and extended hospital stays ($\geq$30 days: $aOR = 0.288, p = 0.001$) were associated with lower odds of in-hospital death, longer hospital stays were inversely associated with in-hospital mortality; however, this finding should be interpreted cautiously because length of stay is partly determined by survival time. Hospitalization and mortality burdens remained stable, driven by acute early-stay mortality and public sector concentration. Coding accuracy improved over time, providing a critical baseline to guide regional neuro-oncological policy and resource distribution**.**

Keywords: Central Nervous System Cancers, Inpatients, Risk Factors, Length of Stay, Health Equity.

Introduction

Malignant brain tumors comprise a heterogeneous group of central nervous system neoplasms characterized by infiltrative growth, a high burden of morbidity, and elevated mortality. Although their incidence is lower than that of many other cancer types, they represent a major cause of disability and death because of their anatomical location, biological behavior, and limited response to available therapeutic interventions [1] . These neoplasms include both primary tumors and metastatic lesions, with high-grade gliomas being the most common primary malignant brain tumors. Early diagnosis and advances in histopathological and molecular characterization have improved tumor classification and clinical management; however, survival remains limited in most cases [2].

These neoplasms include both primary tumors and metastatic lesions, with high-grade gliomas being the most common primary malignant brain tumors. Early diagnosis and advances in histopathological and molecular characterization have improved tumor classification and clinical management; however, survival remains limited in most cases [3], in this context, epidemiological and clinical research is essential to improve the understanding of their biological behavior and to guide strategies for prevention, diagnosis, and treatment.

Globally, malignant brain tumors represent a relatively uncommon but highly lethal group of cancers, imposing a disproportionate burden on health systems due to their poor prognosis and substantial neurological sequelae. According to GLOBOCAN 2022, approximately 321,700 new cases and 248,500 deaths from brain and central nervous system cancers were estimated worldwide, highlighting their high case-fatality despite their comparatively low incidence [4]. The geographic distribution of malignant brain tumors exhibits substantial regional variation, influenced by differences in population structure, diagnostic capacity, cancer registration systems, environmental exposures, and access to specialized neuro-oncology care. These disparities highlight the importance of strengthening epidemiological surveillance and improving equitable access to timely diagnosis and treatment [5].

In Latin America and the Caribbean, malignant brain tumors represent a significant public health challenge because of their high lethality and unequal access to specialized diagnostic and treatment services. According to GLOBOCAN 2022, approximately 27,000 new cases and more than 22,500 deaths from brain and central nervous system cancers were estimated across the region [6]. Incidence and mortality vary considerably among countries, reflecting differences in

population-based cancer registration, diagnostic capacity, and access to specialized oncology care. These disparities highlight the need to strengthen epidemiological surveillance and regional cancer control strategies [7].

In Ecuador, malignant brain tumors are relatively uncommon neoplasms but remain associated with substantial mortality and disability because of their aggressive clinical behavior. According to GLOBOCAN 2024 estimates, approximately 561 new cases of brain and central nervous system cancers and 463 related deaths were recorded in Ecuador [8]. Despite their lower incidence compared with other malignancies, diagnosis frequently occurs at advanced stages, and access to specialized neuro-oncology services remains uneven across the country. These challenges highlight the need to strengthen epidemiological surveillance and improve timely diagnosis and treatment strategies [9].

Accordingly, this study aimed to characterize the hospitalization burden of malignant brain tumors in Ecuador from 2015 to 2024 by examining age-adjusted hospitalization and in-hospital mortality rates, geographic distribution, anatomical patterns, and sociodemographic and institutional factors associated with hospital outcomes..

Methods

Study Design and Data Source

We performed a retrospective observational study based on hospital discharge records collected by INEC between January 1, 2015, and December 31, 2024. The INEC database aggregates comprehensive inpatient admissions across public, private, and social security healthcare institutions spanning all 24 provinces of Ecuador. This manuscript was prepared following the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) guidelines to ensure transparent and reproducible methodology.

Population and Anatomical Classification

The study population comprised all hospital discharge records with a primary discharge diagnosis of a malignant brain neoplasm, identified using International Classification of Diseases, Tenth Revision (ICD-10) codes C71.0 through C71.9. Each record represented a hospital admission rather than a unique individual. Therefore, the unit of analysis was the hospitalization, and repeated admissions of the same individual could not be identified across the study period:

- Supratentorial (Hemispheric): C71.0 (cerebrum), C71.1 (frontal lobe), C71.2 (temporal lobe), C71.3 (parietal lobe), and C71.4 (occipital lobe).
- Infratentorial and Midline: C71.5 (brain ventricles), C71.6 (cerebellum), and C71.7 (brainstem).
- Overlapping Lesions: C71.8 (overlapping malignant lesions of the brain).
- Unspecified / Other: C71.9 (malignant neoplasm of the brain, unspecified).

Variables and Measurements

The primary dependent variable for the binary outcomes was patient survival at discharge, recorded as a binary variable (0 = Alive, 1 = Deceased). For the multinomial frameworks, the dependent outcome was the anatomical tumor category.

Independent covariates encompassed sociodemographic, clinical, and institutional parameters:

- Demographic characteristics: Age group (0–17, 18–44, 45–64, 65–79, and ≥80 years), sex (male, female), and self-reported ethnicity (Mestizo, Afro-Ecuadorian/Afro-descendant, White, Indigenous, Montubio, and Other).
- Geographic and structural factors: Area of residence (urban, rural) and health center sector (public vs. private).
- Clinical and temporal variables: Length of hospital stay (1–3, 4–14, 15–30, and ≥30 days) and calendar year of discharge (2015–2024).

Statistical Analysis

Categorical variables were summarized using absolute frequencies (n) and proportions (%). Group differences across survival status (Dead vs. Alive) and anatomical tumor groups were evaluated using Pearson's chi-square (χ^2) test or Fisher's exact test when expected cell counts were below 5 in more than 20% of the cross-tabulation cells.

To compare disease burden across time while accounting for differences in population age structure, Age-Adjusted Hospitalization Rates (AAHR) and Age-Adjusted In-Hospital Mortality Rates (AAIMR) per 100,000 population were calculated using the direct standardization method. Age-specific hospitalization and in-hospital mortality rates were first calculated for the following age groups: 0–17, 18–44, 45–64, 65–79, and ≥80 years. For each age group, the annual age-specific rate was multiplied by the corresponding proportion of the standard population, defined according to

the age distribution of the 2022 Ecuadorian National Population Census. The resulting weighted age-specific rates were then summed to obtain the annual age-adjusted rate. The same procedure was applied separately to hospitalization and in-hospital mortality outcomes and to each ICD-10 anatomical tumor category. Rates were expressed per 100,000 population.

$$AAHR = \sum_{i=1}^{k}\left(\frac{H_i}{P_i}\times 100.000\right)\omega_i$$

For mortality

$$AAIMR = \sum_{i=1}^{k}\left(\frac{D_i}{P_i}\times 100.000\right)\omega_i$$

where **i** denotes the age group, H_i represents the number of hospitalizations, D_i the number of in-hospital deaths, P_i the corresponding population at risk, and ω_i the proportion of age group **i** in the 2022 Ecuadorian Census standard population.

Two distinct mortality measures were evaluated and should not be interpreted interchangeably. In-hospital case-fatality proportion was calculated as the number of hospital admissions resulting in death divided by the total number of hospital admissions with a primary diagnosis of malignant brain tumor, expressed as a percentage. In contrast, the Age-Adjusted In-Hospital Mortality Rate (AAIMR) represents the number of in-hospital deaths relative to the underlying Ecuadorian population and was expressed per 100,000 population after direct age standardization using the 2022 Ecuadorian Census population as the standard.

$$In - hospital\ case - fatality\ proportion = \frac{In-hospital\ deaths}{Hospital\ admissions} \times 100$$

Multivariate modeling was conducted in two stages:

1. Binary Logistic Regression: Evaluated independent risk factors for in-hospital mortality. Adjusted Odds Ratios (aOR) and 95% Confidence Intervals (95% CI) were calculated, adjusting for age group, sex, ethnicity, area of residence, health sector, ICD-10 anatomical group, length of stay, and year of discharge.
2. Multinomial Logistic Regression: Evaluated sociodemographic and institutional predictors of specific anatomical locations (Infratentorial/Midline, Overlapping Lesions, and Supratentorial). The "Unspecified / Other" group served as the baseline reference category due to its unordered

categorical nature. Results are presented as Adjusted Relative Risk Ratios (aRRR) with 95% CIs.

Covariate selection was driven by data availability in the INEC database and informed by established literature on social determinants in neuro-oncology. All statistical tests were two-tailed, with significance defined at $p < 0.05$. Data processing and modeling were conducted using IBM SPSS Statistics (Version 25.0, Armonk, NY, USA).

Ethics Statement

This study analyzed publicly available, secondary administrative registries managed by the National Institute of Statistics and Censuses (INEC) of Ecuador. The dataset is fully anonymized and stripped of all personal identifiers prior to release. Consequently, the study presented no potential risk to participants, and informed consent was not required in accordance with international ethical standards for secondary data research and national regulations overseen by the Human Research Ethics Committee (CEISH-INSPI).

Results

Between 2015 and 2024, a total of 8,249 hospital admissions with a primary diagnosis of malignant brain tumor were recorded in Ecuador, of which 793 resulted in in-hospital death, corresponding to an overall in-hospital case-fatality proportion of 9.61% ($n = 793$). Bivariate analysis revealed significant disparities in mortality across age groups ($p = 0.001$), where older adults aged 45–64 years concentrated the highest proportion of fatal cases (31.0%), whereas younger cohorts (0–44 years) accounted for over half of all admissions resulting in live discharge (56.5%). Geographic and institutional barriers significantly influenced outcomes; rural residents (15.6% vs. 13.1%, $p = 0.044$) and patients treated in public sector facilities (65.3% vs. 59.1%, $p = 0.001$) experienced higher proportions of in-hospital deaths compared to their urban and private-sector counterparts. Furthermore, prolonged hospitalization was strongly associated with mortality ($p = 0.001$), with extended stays of ≥30 days being more than twice as frequent among admissions resulting in in-hospital death compared to survivors (17.7% vs. 6.9%). Neither sex ($p = 0.548$) nor ICD-10 anatomical tumor location ($p = 0.139$) demonstrated statistically significant differences in crude in-hospital survival (Table 1).

Table 1. Sociodemographic, institutional, and clinical characteristics of hospitalized patients with malignant brain tumors in Ecuador stratified by in-hospital vital outcome.

Variables	Dead (n=793)		Alive (n=7456)		p Value
	n	%	n	%	
Age groups					**0.001**
0-17 years	178	22.4%	2124	28.5%	
18-44 years	171	21.6%	2089	28.0%	
45-64 years	246	31.0%	1975	26.5%	
65-79 years	166	20.9%	1079	14.5%	
≥80 years	32	4.0%	189	2.5%	
Sex					0.548
Male	436	55.0%	4016	53.9%	
Female	357	45.0%	3440	46.1%	
Ethnicity					**0.014**
Afro-Ecuadorian/ Afro-descendant	5	0.6%	29	0.4%	
White	39	4.9%	274	3.7%	
Indigenous	10	1.3%	65	0.9%	
Mestizo	629	79.3%	6038	81.0%	
Montubio	5	0.6%	11	0.1%	
Other	105	13.2%	1039	13.9%	
Area of residence					**0.044**
Rural	124	15.6%	975	13.1%	
Urban	669	84.4%	6481	86.9%	
Sector of health center					**0.001**
Private	275	34.7%	3051	40.9%	
Public	518	65.3%	4405	59.1%	
Categories (ICD-10)					0.139
Infratentorial and Midline	105	13.2%	1080	14.5%	
Overlapping Lesions	6	0.8%	87	1.2%	
Unspecified / Other	372	46.9%	3198	42.9%	
Supratentorial (Hemispheric)	310	39.1%	3091	41.5%	
Days of Hospitalization					**0.001**
1-3 days	207	26.1%	2480	33.3%	
4-14 days	287	36.2%	3234	43.4%	
15-30 days	159	20.1%	1227	16.5%	
≥30 days	140	17.7%	515	6.9%	
Year of discharge					**0.001**
2015	91	11.5%	922	12.4%	
2016	88	11.1%	894	12.0%	
2017	10	1.3%	21	0.3%	
2018	95	12.0%	949	12.7%	
2019	92	11.6%	1004	13.5%	
2020	75	9.5%	857	11.5%	
2021	83	10.5%	976	13.1%	
2022	91	11.5%	911	12.2%	
2023	83	10.5%	0	0.0%	
2024	85	10.7%	895	9.5%	

p Values were calculated with the X^2 test or Fisher's exact for comparing categorical variables.
Values in bold are of statistical significance.
*Fisher's exact test was used if >20% of the cells had expected frequencies of less than five.

Anatomical localization of malignant brain neoplasms demonstrated highly significant associations with age, ethnicity, health sector, length of stay, and discharge year ($p = 0.001$ for

all). Infratentorial and midline tumors were heavily concentrated among pediatric patients (43.2% aged 0–17 years, $n = 512$), whereas supratentorial and unspecified lesions predominated in adult populations aged 18–64 years (52.2% and 58.7%, respectively). Healthcare sector utilization varied considerably by tumor location; while public facilities treated 67.7% ($n = 2,302$) of supratentorial cases and 77.4% ($n = 72$) of overlapping lesions, infratentorial tumors were equally managed across private (50.4%) and public (49.6%) centers. Extended hospital stays (≥30 days) were most common in supratentorial (8.9%) and infratentorial (8.6%) cases. Patient sex ($p = 0.216$) and area of residence ($p = 0.141$) did not differ significantly across anatomical sites (Table 2).

Table 2. Sociodemographic, institutional, and clinical characteristics of hospitalized cases with primary malignant brain tumors in Ecuador stratified by ICD-10 anatomical tumor location.

Variable	Infratentorial and Midline (n=1185)		Overlapping Lesions (n=93)	
	n	%	n	%
Age groups				
0-17 years	512	43.2%	11	11.8%
18-44 years	353	29.8%	37	39.8%
45-64 years	197	16.6%	26	28.0%
65-79 years	100	8.4%	15	16.1%
≥80 years	23	1.9%	4	4.3%
Sex				
Male	664	56.0%	56	60.2%
Female	521	44.0%	37	39.8%
Ethnicity				
Afro-Ecuadorian/ Afro-descendant	3	0.3%	1	1.1%
White	15	1.3%	6	6.5%
Indigenous	8	0.7%	0	0.0%
Mestizo	1033	87.2%	72	77.4%
Montubio	4	0.3%	0	0.0%
Other	122	10.3%	14	15.1%
Area of residence				
Rural	168	14.2%	12	12.9%
Urban	1017	85.8%	81	87.1%
Sector of health center				
Private	597	50.4%	21	22.6%
Public	588	49.6%	72	77.4%
Days of Hospitalization				
1-3 days	452	38.1%	23	24.7%
4-14 days	463	39.1%	49	52.7%
15-30 days	168	14.2%	15	16.1%
≥30 days	102	8.6%	6	6.5%
Year of discharge				
2015	102	8.6%	18	19.4%
2016	94	7.9%	2	2.2%
2017	1	0.1%	0	0.0%

2018	147	12.4%	9	9.7%
2019	183	15.4%	8	8.6%
2020	150	12.7%	8	8.6%
2021	172	14.5%	14	15.1%
2022	189	15.9%	17	18.3%
2023	11	0.9%	0	0.0%
2024	136	11.5%	17	18.3%

Continuing table

Variable	Unspecified / Other (n=3570)		Supratentorial (Hemispheric) (n=3401)		p Value
	n	%	n	%	
Age groups					**0.001**
0-17 years	832	23.3%	947	27.8%	
18-44 years	1034	29.0%	836	24.6%	
45-64 years	1060	29.7%	938	27.6%	
65-79 years	549	15.4%	581	17.1%	
≥80 years	95	2.7%	99	2.9%	
Sex					0.216
Male	1925	53.9%	1807	53.1%	
Female	1645	46.1%	1594	46.9%	
Ethnicity					**0.001**
Afro-Ecuadorian/ Afro-descendant	20	0.6%	10	0.3%	
White	171	4.8%	121	3.6%	
Indigenous	36	1.0%	31	0.9%	
Mestizo	2803	78.5%	2759	81.1%	
Montubio	5	0.1%	7	0.2%	
Other	535	15.0%	473	13.9%	
Area of residence					0.141
Rural	501	14.0%	418	12.3%	
Urban	3069	86.0%	2983	87.7%	
Sector of health center					**0.001**
Private	1609	45.1%	1099	32.3%	
Public	1961	54.9%	2302	67.7%	
Days of Hospitalization					**0.001**
1-3 days	1149	32.2%	1063	31.3%	
4-14 days	1582	44.3%	1427	42.0%	
15-30 days	596	16.7%	607	17.8%	
≥30 days	243	6.8%	304	8.9%	
Year of discharge					**0.001**
2015	518	14.5%	375	11.0%	
2016	486	13.6%	400	11.8%	
2017	11	0.3%	19	0.6%	
2018	545	15.3%	343	10.1%	
2019	426	11.9%	479	14.1%	
2020	380	10.6%	394	11.6%	
2021	419	11.7%	454	13.3%	
2022	367	10.3%	429	12.6%	
2023	39	1.1%	33	1.0%	
2024	379	10.6%	475	14.0%	

p Values were calculated with the X^2 test for comparing categorical variables.
Values in bold are of statistical significance.

Between 2015 and 2024, the overall 10-year mean Age-Adjusted Hospitalization Rate (AAHR) for primary malignant brain tumors in Ecuador was 4.859 per 100,000 population.

Excluding years with unusually low recorded hospitalization counts in 2017 (0.185 per 100,000) and 2023 (0.485 per 100,000), admission rates remained stable between 5.322 and 6.393 per 100,000 inhabitants, reaching a peak in 2019 (6.393 per 100,000) and dropping in 2020 (5.322 per 100,000) coinciding with the first year of the COVID-19 pandemic healthcare disruptions. Unspecified neoplasms recorded the highest 10-year mean AAHR (2.086 per 100,000), followed closely by supratentorial lesions (1.996 per 100,000). Notably, the recorded frequency of unspecified anatomical coding decreased by 2024, as the unspecified rate dropped to 0.000 per 100,000 while supratentorial admissions reached a study period high of 2.747 per 100,000.

The 10-year mean Age-Adjusted In-Hospital Mortality Rate (AAIMR) stood at 0.461 per 100,000 population, remaining remarkably stable across non-anomalous years with peak rates observed in 2015 and 2018 (0.561 per 100,000). By anatomical localization, unspecified tumors (0.220 per 100,000) and supratentorial lesions (0.182 per 100,000) accounted for over 87% of the total standardized mortality burden nationwide, whereas infratentorial/midline lesions (0.062 per 100,000) and overlapping tumors (0.004 per 100,000) contributed minimally to overall population-level mortality (Table 3).

Table 3. Trends in age-adjusted hospitalization rates (AAHR) and age-adjusted in-hospital mortality rates (AAIMR) per 100,000 population for primary malignant brain tumors in Ecuador, stratified by ICD-10 anatomical location.

Years	Age adjusted hospitalization rates				
	Total Cases	Overlapping Lesions	Infratentorial and Midline	Supratentorial (Hemispheric)	Unspecified / Other
2015	6.249	0.111	0.629	2.313	3.195
2016	5.965	0.012	0.571	2.430	2.952
2017	0.185	0.000	0.006	0.114	0.066
2018	6.164	0.053	0.868	2.025	3.218
2019	6.393	0.047	1.067	2.794	2.485
2020	5.322	0.046	0.857	2.250	2.170
2021	5.967	0.079	0.969	2.558	2.361
2022	6.033	0.100	1.116	2.533	2.167
2023	0.485	0.000	0.064	0.193	0.228
2024	3.631	0.098	0.786	2.747	0.000
Mean rate between 2015 and 2024	4.859	0.055	0.693	1.996	2.086

Continuing table

Years	Age adjusted in-hospital mortality rates				
	Total Cases	Overlapping Lesions	Infratentorial and Midline	Supratentorial (Hemispheric)	Unspecified / Other
2015	0.561	0.006	0.068	0.204	0.284
2016	0.534	0.000	0.055	0.146	0.334
2017	0.060	0.000	0.006	0.036	0.018

2018	0.561	0.000	0.047	0.171	0.342
2019	0.537	0.000	0.076	0.204	0.257
2020	0.428	0.000	0.051	0.188	0.188
2021	0.468	0.000	0.096	0.225	0.146
2022	0.537	0.012	0.089	0.230	0.207
2023	0.485	0.000	0.064	0.193	0.228
2024	0.492	0.017	0.064	0.220	0.191
Mean rate between 2015 and 2024	0.461	0.004	0.062	0.182	0.220

Note: Were standardized using the direct method based on the 2022 Ecuadorian Census population.

Age-adjusted hospitalization and in-hospital mortality rates for malignant brain tumors showed substantial temporal variation in Ecuador between 2015 and 2024. Hospitalization rates remained relatively stable at approximately 5–6 cases per 100,000 population during most years, reaching their highest value in 2019 (6.393 per 100,000), while pronounced reductions were observed in 2017 (0.185) and 2023 (0.485). Supratentorial (hemispheric) and unspecified/other tumors accounted for the largest components of the hospitalization rates across most of the study period. Age-adjusted in-hospital mortality rates were considerably lower, generally ranging from 0.428 to 0.561 per 100,000 in years other than 2017, with the lowest value recorded in 2017 (0.006). Overall, the temporal patterns indicate fluctuations in both hospitalization and mortality rates, with particularly atypical values in 2017 and 2023 that warrant cautious interpretation.

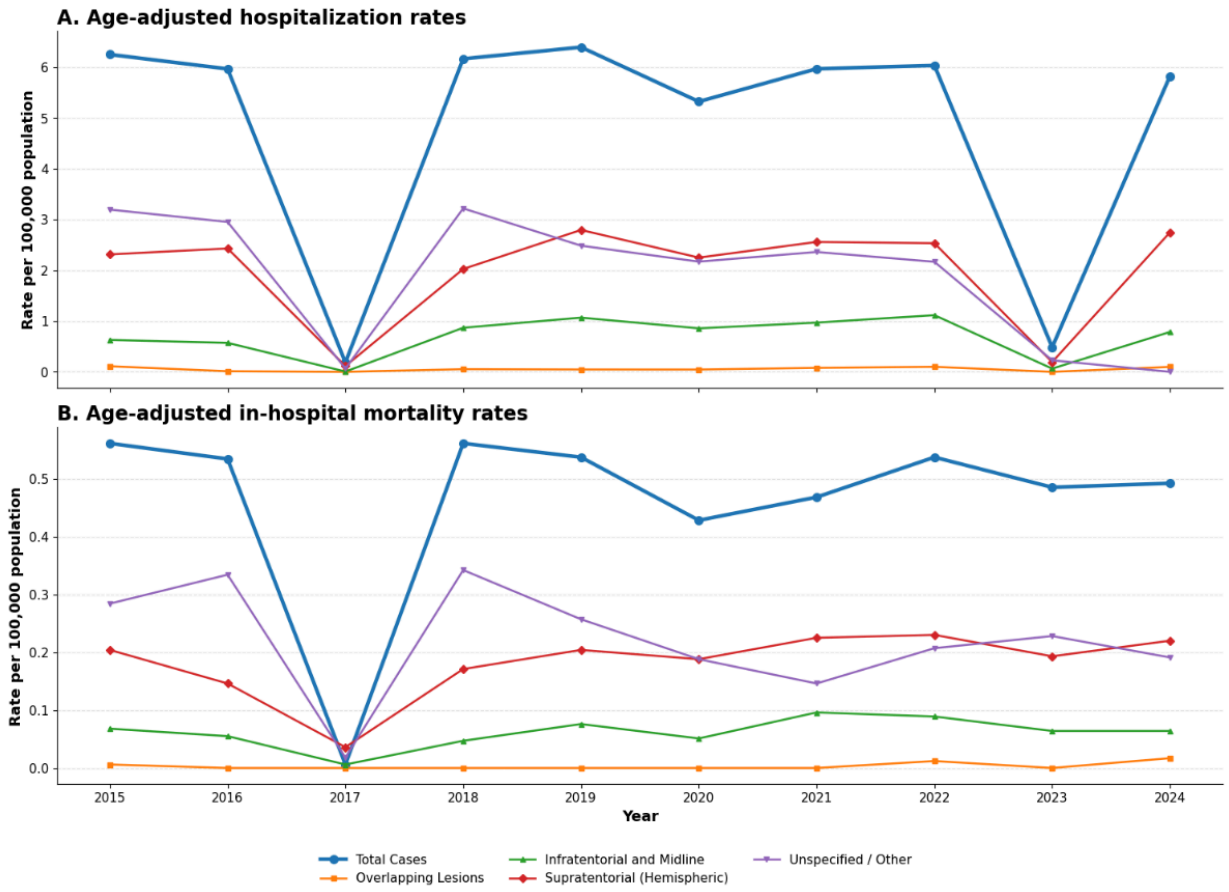


Figure 1. Temporal trends in age-adjusted hospitalization and in-hospital mortality rates for malignant brain tumors in Ecuador, 2015–2024.

After controlling for sociodemographic, spatial, and clinical covariates, multivariate binary logistic regression revealed that age group and length of hospital stay were the primary independent predictors of in-hospital survival. Compared to patients aged 0–17 years, older age cohorts demonstrated significantly lower adjusted odds of in-hospital death, ranging from $aOR = 0.671$ ($95\%\ CI: 0.536 -- 0.839$, $p = 0.001$) in individuals aged 45–64 years to $aOR = 0.457$ ($95\%\ CI: 0.296 -- 0.707$, $p = 0.001$) in those aged ≥80 years. Length of hospital stay was significantly associated with in-hospital mortality. Compared with admissions lasting 1–3 days, stays of 15–30 days (aOR = 0.710, 95% CI: 0.556–0.907; p = 0.006) and ≥30 days (aOR = 0.288, 95% CI: 0.222–0.372; p < 0.001) were associated with lower adjusted odds of in-hospital death. These associations should not be interpreted as evidence of a protective effect of prolonged hospitalization, because length of stay is partly determined by survival time during hospitalization. (Table 4).

Table 4. Table 4. Multivariate binary logistic regression model assessing sociodemographic, clinical, and structural predictors of in-hospital mortality among patients with primary malignant brain tumors in Ecuador.

Variables	aOR	(95% CI) Lower	Upper	p Value
Age groups				
0-17 years		Reference		
18-44 years	1.037	0.817	1.316	0.767
45-64 years	0.671	0.536	0.839	**0.001**
65-79 years	0.523	0.408	0.671	**0.001**
≥80 years	0.457	0.296	0.707	**0.001**
Sex				
Male		Reference		
Female	1.052	0.899	1.232	0.526
Ethnicity				
Afro-Ecuadorian/ Afro-descendant		Reference		
White	0.859	0.242	3.043	0.813
Indigenous	0.938	0.230	3.826	0.929
Mestizo	1.077	0.320	3.629	0.905
Montubio	0.192	0.036	1.011	0.052
Other	1.615	0.470	5.553	0.446
Area of residence				
Rural		Reference		
Urban	1.203	0.964	1.501	0.103
Sector of health center				
Private		Reference		
Public	0.876	0.731	1.048	0.148
Categories (ICD-10)				
Infratentorial y Línea Media		Reference		
Lesiones Sobrepuestas	1.414	0.596	3.354	0.432
No Especificado / Otros	0.917	0.716	1.176	0.496
Supratentorial (Hemisférico)	1.126	0.876	1.448	0.355
Days of Hospitalization				
1-3 days		Reference		
4-14 days	0.935	0.764	1.144	0.515
15-30 days	0.710	0.556	0.907	**0.006**
≥30 days	0.288	0.222	0.372	**0.001**
Year of discharge				
2015		Reference		
2016	0.993	0.724	1.360	0.963
2017	0.321	0.139	0.739	**0.008**
2018	1.065	0.782	1.449	0.690
2019	1.135	0.832	1.548	0.425
2020	1.202	0.866	1.667	0.272
2021	1.118	0.813	1.536	0.493
2022	0.939	0.687	1.283	0.691
2023	0.001	0.001	-	0.996
2024	1.088	0.792	1.494	0.604

Odds ratios adjusted for age group, sex, ethnicity, area of residence, health center sector, schizophrenia classification, and year of discharge.
Values in bold are statistically significant.

Multinomial logistic regression identified age, ethnicity, area of residence, health sector, and length of stay as independent predictors of anatomical tumor site relative to unspecified neoplasms. Pediatric age (0–17 years) significantly increased the relative risk of infratentorial and midline tumor diagnoses ($aRRR = 2.424$, 95% CI: $1.506 -- 3.902$, $p = 0.001$). Mestizo ethnicity was associated with higher relative risk for both supratentorial ($aRRR = 1.386$, $p = 0.001$) and infratentorial ($aRRR = 1.331$, $p = 0.012$) diagnoses. Institutional determinants were notable: private healthcare facilities recorded significantly lower relative odds for supratentorial ($aRRR = 0.460$, $p = 0.001$) and overlapping lesions ($aRRR = 0.289$, $p = 0.001$) compared to public centers. Rural residence was associated with reduced relative odds of supratentorial tumors ($aRRR = 0.754$, $p = 0.001$). Neither sex nor patient age showed significant associations with supratentorial localization relative to unspecified lesions (Table 5).

Table 5. Multinomial logistic regression model evaluating sociodemographic, structural, and temporal factors associated with specific ICD-10 anatomical tumor locations relative to the "Unspecified / Other" baseline category.

Variables	Infratentorial and Midline				Overlapping Lesions			
	aRRR	(95% CI)		p Value	aRRR	(95% CI)		p Value
		Lower	Upper			Lower	Upper	
Age groups								
0-17 years	2.424	1.506	3.902	**0.001**	0.437	0.133	1.432	0.172
18-44 years	1.393	0.865	2.242	0.173	0.907	0.313	2.628	0.857
45-64 years	0.780	0.481	1.267	0.316	0.620	0.210	1.832	0.387
65-79 years	0.781	0.470	1.298	0.340	0.694	0.223	2.158	0.528
≥80 years	Reference				Reference			
Sex								
Male	1.121	0.978	1.284	0.100	1.258	0.822	1.924	0.290
Female	Reference				Reference			
Ethnicity								
Afro-Ecuadorian/ Afro-descendant	0.488	0.140	1.698	0.259	3.983	0.480	33.071	0.201
White	0.443	0.250	0.785	**0.005**	2.691	0.974	7.433	0.056
Indigenous	0.851	0.380	1.907	0.695	-	-	-	0.997
Mestizo	1.331	1.064	1.666	**0.012**	1.496	0.820	2.729	0.190
Montubio	2.657	0.681	10.372	0.160	-	-	-	-
Other	Reference				Reference			
Area of residence								
Rural	0.954	0.785	1.160	0.636	0.744	0.399	1.389	0.353
Urban	Reference				Reference			
Sector of health center								
Private	0.865	0.744	1.005	0.058	0.289	0.171	0.489	**0.001**
Public	Reference				Reference			
Days of Hospitalization								
1-3 days	0.883	0.675	1.155	0.364	1.055	0.418	2.664	0.910
4-14 days	0.713	0.548	0.926	**0.011**	1.289	0.541	3.071	0.566
15-30 days	0.766	0.571	1.030	0.077	0.869	0.330	2.284	0.775
≥30 days	Reference				Reference			

Year of discharge								
2015	0.508	0.378	0.682	**0.001**	0.660	0.332	1.312	0.236
2016	0.475	0.351	0.644	**0.001**	0.067	**0.015**	0.298	**0.001**
2017	0.223	0.028	1.766	0.155	-	**0.001**	-	0.998
2018	0.754	0.573	0.991	**0.043**	0.275	0.119	0.634	**0.002**
2019	1.178	0.901	1.540	0.230	0.335	0.141	0.793	**0.013**
2020	0.991	0.751	1.309	0.950	0.427	0.180	1.009	0.052
2021	1.025	0.783	1.341	0.857	0.739	0.358	1.529	0.415
2022	1.270	0.972	1.660	0.080	1.113	0.556	2.226	0.763
2023	0.752	0.370	1.529	0.431	-	-	-	-
2024	Reference				Reference			

Continuing table

Variables	Supratentorial (Hemispheric)			
	aRRR	(95% CI)		p Value
		Lower	Upper	
Age groups				
0-17 years	1.265	0.932	1.718	0.132
18-44 years	0.814	0.601	1.102	0.183
45-64 years	0.899	0.665	1.216	0.491
65-79 years	1.089	0.797	1.488	0.593
≥80 years	Reference			
Sex				
Male	0.991	0.900	1.092	0.856
Female	Reference			
Ethnicity				
Afro-Ecuadorian/ Afro-descendant	0.782	0.355	1.724	0.543
White	0.984	0.749	1.292	0.907
Indigenous	1.041	0.624	1.735	0.879
Mestizo	1.386	1.199	1.601	**0.001**
Montubio	2.251	0.693	7.312	0.177
Other	Reference			
Area of residence				
Rural	0.754	0.653	0.871	**0.001**
Urban	Reference			
Sector of health center				
Private	0.460	0.412	0.513	**0.001**
Public	Reference			
Days of Hospitalization				
1-3 days	0.882	0.725	1.073	0.210
4-14 days	0.799	0.662	0.965	**0.020**
15-30 days	0.842	0.684	1.037	0.105
≥30 days	Reference			
Year of discharge				
2015	0.512	0.422	0.622	**0.001**
2016	0.551	0.453	0.670	**0.001**
2017	1.113	0.515	2.405	0.785
2018	0.450	0.370	0.547	0.001
2019	0.833	0.687	1.009	0.062
2020	0.786	0.644	0.959	**0.018**
2021	0.843	0.695	1.022	0.083
2022	0.941	0.772	1.146	0.543
2023	0.591	0.362	0.967	**0.036**
2024	Reference			

Relative risk ratios adjusted for age groups, sex, ethnicity, area of residence and year of discharge.
Values in bold are of statistical significance.

The spatial distribution of malignant brain tumors showed marked heterogeneity across Ecuadorian provinces during 2015–2024. Pichincha and Guayas consistently concentrated the highest numbers of cases, reaching their maximum values in 2019 (427 cases) and 2016 (449 cases), respectively. Azuay represented another relevant concentration, particularly between 2018 and 2022, whereas Manabí and Loja showed intermediate numbers and variations throughout the study period. In contrast, several Amazonian provinces and Galápagos reported considerably fewer cases. An exceptional reduction in recorded cases was observed in 2017 and 2023 across nearly the entire country, followed by an increase in subsequent years; this pattern should be interpreted with caution, as it may reflect changes in data recording or availability rather than a true decline in disease occurrence. Overall, the maps reveal a persistent concentration of cases mainly in Pichincha, Guayas, and, to a lesser extent, Azuay.

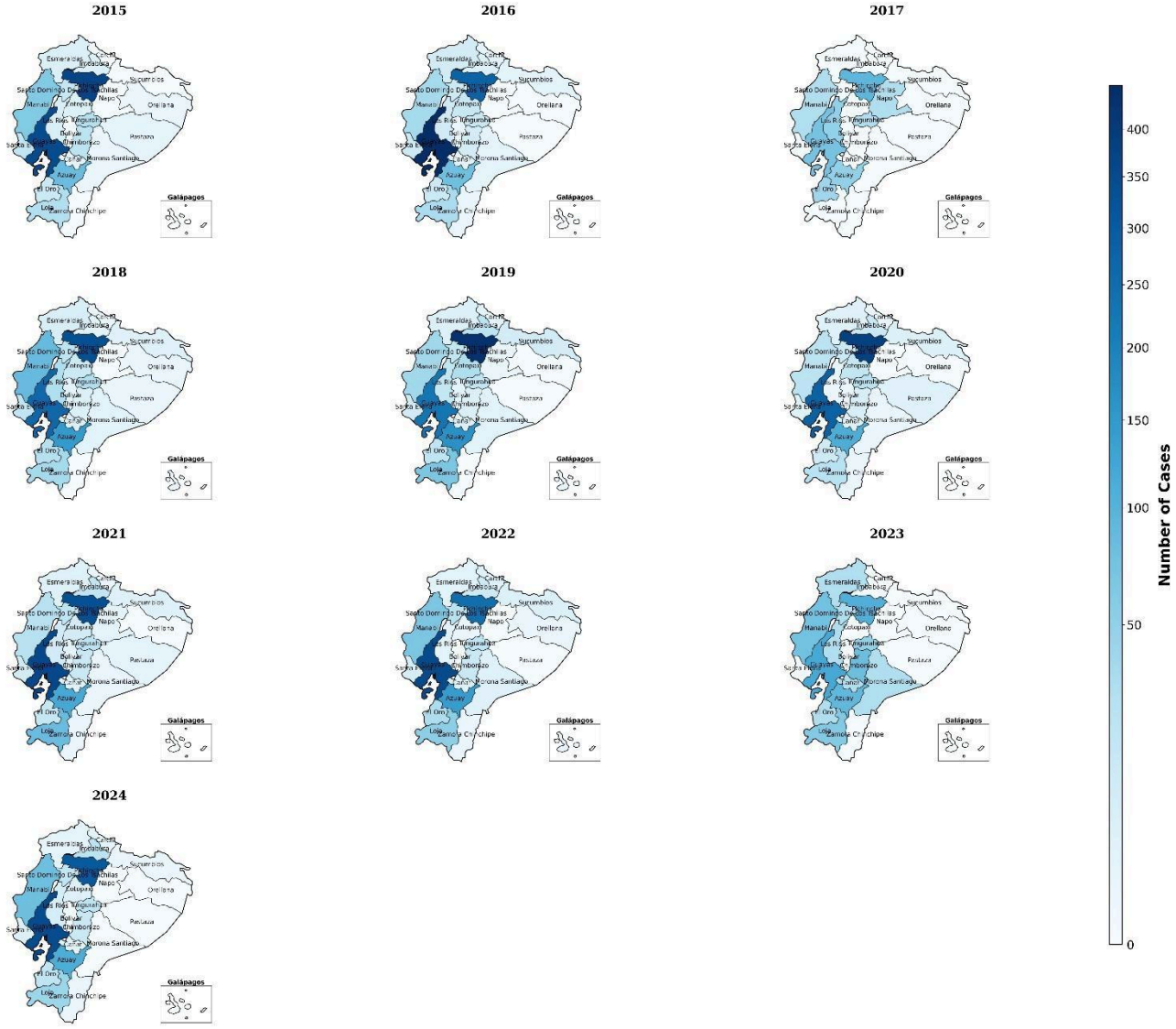


Figure 2. Spatial and temporal distribution of malignant brain tumor cases by province, Ecuador, 2015–2024

Discussion

This study provides the first decade-long (2015–2024) nationwide evaluation of age-adjusted hospitalization (AAHR: 4.859 per 100,000) and in-hospital mortality rates (AAIMR: 0.461 per 100,000) for primary malignant brain tumors in Ecuador [10]. The primary novelty lies in uncovering critical demographic and administrative transitions: pediatric patients (≤17 years) concentrated nearly half of all infratentorial/midline neoplasms (43.2%, $aRRR = 2.424$), whereas adult cohorts predominated in supratentorial sites [11]. Furthermore, longitudinal tracking revealed a dramatic structural improvement in institutional reporting, marked by the total resolution of unspecified tumor coding ($AAHR = 0.000$) alongside a historical peak in supratentorial diagnoses ($AAHR = 2.747$) by 2024 [12]. Extended hospital stays (≥30 days) were associated with lower adjusted odds of in-hospital death (aOR = 0.288). However, this finding should be interpreted cautiously because length of stay is inherently conditioned by survival during hospitalization; patients must survive long enough to enter the prolonged-stay category. Therefore, this association should not be interpreted as evidence that prolonged hospitalization confers a protective effect against in-hospital mortality.

While male predominance was consistently observed across all anatomical sites (53.1% to 60.2%), sex was not an independent predictor of mortality or tumor localization [13]. During the first year of the COVID-19 pandemic (2020), total hospitalization rates dropped sharply to 5.322 per 100,000 (down from 6.393 in 2019), reflecting severe health system strain, deferred elective surgical admissions, and delayed neuroimaging access across Ecuador [14]. However, in-hospital mortality remained relatively stable ($AAIMR = 0.428$), suggesting that emergency care for critically ill neuro-oncological patients was prioritized [15]. Moreover, the extreme rate reductions in 2017 (0.185) and 2023 (0.485) underscore persistent administrative registry capture disruptions rather than true epidemiological shifts in population burden.

Ecuador's overall hospitalization metrics align closely with rates reported across Latin America and other developing regions (e.g., Brazil, Mexico, Colombia, Peru, India, and China), where rates generally range between 3.5 and 6.0 per 100,000 population [16]. These figures contrast sharply with high-income settings, such as North America and Western Europe, which consistently document higher rates (7.0 to 11.5 per 100,000) due to ubiquitous 3T MRI accessibility, integrated electronic registries, and higher diagnostic rates in geriatric populations [17] . In

low-to-middle-income countries (LMICs), lower observed rates are frequently influenced by underdiagnosis, diagnostic infrastructure gaps in rural zones, and lower inpatient admission rates among elderly terminal patients [18]. The heavy reliance on public facilities for complex cases in Ecuador (67.7% for supratentorial lesions) mirrors broader structural patterns across Latin American public health systems.

This investigation has several limitations inherent to secondary administrative health datasets [19]. Because the INEC administrative database does not allow longitudinal identification of unique individuals, the unit of analysis was the hospitalization rather than the patient; therefore, repeated admissions of the same individual may be represented as separate records. Hospital discharge registries (INEC) are susceptible to misclassification, coding heterogeneity across clinical centers, and single-year capture anomalies (2017 and 2023) [20]. Furthermore, ICD-10 coding lacks integrated WHO molecular and histopathological granularity (e.g., IDH mutation status, MGMT methylation, or 1p/19q codeletion) [21]. Finally, the dataset strictly captures in-hospital deaths, thereby excluding out-of-hospital or home-based palliative mortality.

Despite these constraints, the study's principal strengths lie in its comprehensive 10-year nationwide scope, standardization against the updated 2022 National Census, and rigorous multivariate adjustments (aOR and $aRRR$) controlling for demographic, institutional, and clinical confounders [22]. By documenting concrete advances in national diagnostic coding precision alongside persistent public-private healthcare access disparities, these findings establish a crucial epidemiological baseline to inform future neuro-oncological health policies [23]. Ultimately, these epidemiological insights provide actionable data for resource allocation, neuro-oncological referral pathways, and specialized surgical infrastructure development across the Andean region [24].

Conclusion

This 10-year nationwide study provides the first comprehensive population-based baseline of age-adjusted hospitalization and in-hospital mortality rates for primary malignant brain tumors in Ecuador. Over the past decade, overall hospitalization and mortality burden remained relatively stable across non-anomalous years, with significant demographic, institutional, and clinical disparities shaping patient outcomes. Pediatric patients concentrated a disproportionate burden of infratentorial and midline neoplasms, whereas supratentorial tumors predominated in adults and older cohorts. Multivariate modeling identified significant associations between age group, length of hospital stays, and in-hospital mortality. The inverse association observed for prolonged hospital

stays should be interpreted cautiously because length of stay is conditioned by survival during hospitalization and may therefore be affected by reverse causation and time-related bias.

Furthermore, institutional analysis highlights significant disparities between public and private healthcare sectors, with public tertiary facilities managing the vast majority of complex supratentorial and overlapping lesions. Notably, temporal trends demonstrate a marked structural advance in national health registries, characterized by the progressive elimination of unspecified anatomical coding and a concurrent rise in diagnostic precision by 2024. These findings underscore the urgent need for targeted public health strategies, equitable distribution of specialized neuro-oncological resources, and streamlined referral networks to mitigate early in-hospital mortality and reduce care barriers across the Andean region.

REFERENCES

[1] Miller KD, Ostrom QT, Kruchko C, Patil N, Tihan T, Cioffi G, et al. Brain and other central nervous system tumor statistics, 2021. CA: A Cancer Journal for Clinicians 2021;71:381–406. https://doi.org/10.3322/CAAC.21693.

[2] Davalan W, Alkins R. Prognostic and predictive determinants in high-grade gliomas: integrating tumor-intrinsic biology with patient and system-level factors. Frontiers in Neurology 2025;16:1664458. https://doi.org/10.3389/FNEUR.2025.1664458/XML.

[3] Roda D, Veiga P, Melo JB, Carreira IM, Ribeiro IP. Principles in the Management of Glioblastoma. Genes 2024, Vol 15, Page 501 2024;15:501. https://doi.org/10.3390/GENES15040501.

[4] Kim S, Son Y, Oh J, Kim S, Jang W, Lee S, et al. Global burden of brain and central nervous system cancer in 185 countries, and projections up to 2050: a population-based systematic analysis of GLOBOCAN 2022. Journal of Neuro-Oncology 2025;175:673–85. https://doi.org/10.1007/S11060-025-05164-0/METRICS.

[5] Price M, Ballard C, Benedetti J, Neff C, Cioffi G, Waite KA, et al. CBTRUS Statistical Report: Primary Brain and Other Central Nervous System Tumors Diagnosed in the United States in 2017–2021. Neuro-Oncology 2024;26:vi1–85. https://doi.org/10.1093/NEUONC/NOAE145.

[6] Leite LF, da Conceição LD, Saldanha EF, Menezes S, de Melo AC, Borea R, et al. Cancer Incidence and Mortality Estimates in Latin America and the Caribbean: A Systematic Analysis of the GLOBOCAN 2022. Cancer Research Communications 2025;5:2236–48. https://doi.org/10.1158/2767-9764.CRC-25-0564/767351/AM/CANCER-INCIDENCE-AND-MORTALITY-ESTIMATES-IN-LATIN.

[7] Jemal A, Parkin DM, Bray F. Patterns of Cancer Incidence, Mortality, and Survival. Schottenfeld and Fraumeni Cancer Epidemiology and Prevention, Fourth Edition 2017:107–40. https://doi.org/10.1093/OSO/9780190238667.003.0008.

[8] Global Cancer Observatory n.d. https://gco.iarc.who.int/en (accessed July 20, 2026).

[9] Maldonado-Noboa I, Checa-Ron F, Tarupi-Montenegro W, Moreno-Ocampo A, Noboa-Jaramillo A, Pacheco-Ojeda L, et al. Cáncer en el Ecuador, ¿Qué ha cambiado en las últimas 3 décadas? Metro Ciencia 2025;33:s56–69. https://doi.org/10.47464/METROCIENCIA/VOL33/SUPLEMENTO-1/2025/S56-S69.

[10] Villarreal-Garza C, Aranda-Gutierrez A, Gonzalez-Sanchez DG, Bragança-Xavier C, Negrete-Tobar G, Chavarri-Guerra Y, et al. National cancer control plans in Latin America and the Caribbean: challenges and future directions. Lancet Oncol 2025;26:e320–30.
[11] Organization WH. WHO country stories: delivering for all. World Health Organization; 2023.
[12] Shakir M, Shariq SF, Irshad HA, Khowaja AH, Tahir I, Rae AI, et al. Barriers to neurosurgical care of brain tumors in low-and middle-income countries: a systematic review of the service delivery challenges. World Neurosurg 2024;187:211–22.
[13] Price M, Ballard C, Benedetti J, Neff C, Cioffi G, Waite KA, et al. CBTRUS statistical report: primary brain and other central nervous system tumors diagnosed in the United States in 2017–2021. Neuro Oncol 2024;26:vi1.
[14] Louis DN, Perry A, Wesseling P, Brat DJ, Cree IA, Figarella-Branger D, et al. The 2021 WHO classification of tumors of the central nervous system: a summary. Neuro Oncol 2021;23:1231–51.
[15] Girardi F, Allemani C, Coleman MP. Worldwide trends in survival from common childhood brain tumors: a systematic review. J Glob Oncol 2019;5:1–25.
[16] Sung H, Ferlay J, Siegel RL, Laversanne M, Soerjomataram I, Jemal A, et al. Global cancer statistics 2020: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries. CA Cancer J Clin 2021;71:209–49.
[17] Gupta T, Achari R, Chatterjee A, Chen Z-P, Mehta M, Bouffet E, et al. Comparison of epidemiology and outcomes in neuro-oncology between the east and the west: challenges and opportunities. Clin Oncol 2019;31:539–48.
[18] Ramirez Ferrer E, Zuluaga-Garcia JP, Alzate JD, Mayorga-Corvacho J, Sierra MA, Garzon-Duque MO, et al. Impact of healthcare disparities on the treatment and survival of patients with high-grade gliomas in Colombia: a multicenter inverse probability-weighted cohort analysis. J Neurooncol 2025;175:1067–77.
[19] Beltrán JQ, Soto-Abraham JE, Vidaurreta-Serrano J, Macias LGC, Apo EG, Ogando-Rivas E. Astrocytic tumors in Mexico: An overview of characteristics and prognosis in an open reference center for low-income population. J Neurosci Rural Pract 2017;9:516–21.
[20] Visser O, Ardanaz E, Botta L, Sant M, Tavilla A, Minicozzi P, et al. Survival of adults with primary malignant brain tumours in Europe; Results of the EUROCARE-5 study. Eur J Cancer 2015;51:2231–41.
[21] Bray F, Laversanne M, Sung H, Ferlay J, Siegel RL, Soerjomataram I, et al. Global cancer statistics 2022: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries. CA Cancer J Clin 2024;74:229–63.
[22] Huang J, Chan SC, Lok V, Zhang L, Lin X, Lucero-Prisno DE, et al. Disease burden, risk factors, and trends of primary central nervous system (CNS) cancer: a global study of registries data. Neuro Oncol 2023;25:995–1005.
[23] Pineros M, Sierra MS, Izarzugaza MI, Forman D. Descriptive epidemiology of brain and central nervous system cancers in Central and South America. Cancer Epidemiol 2016;44:S141–9.
[24] Low JT, Ostrom QT, Cioffi G, Neff C, Waite KA, Kruchko C, et al. Primary brain and other central nervous system tumors in the United States (2014-2018): A summary of the CBTRUS statistical report for clinicians. Neurooncol Pract 2022;9:165–82.